

Unsupervised Learning Laboratories

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Final lab

You can use external libraries for linear algebra operations but you are expected to write your own algorithms.

Exercise 1

Implement your own versions of:

- Density peaks clustering (without the "halo" feature)
- Normalized Mutual Information (NMI)
- F-ratio

Using the Aggregation dataset available at <https://cs.joensuu.fi/sipu/datasets/> under Shape sets:

1. Use **your** version of density peaks at several values of the ϵ parameter and choose the one that maximises the NMI with the ground truth (third column).
2. Compute the NMI with the ground truth and the F-ratio test for this assignation.
3. Perform k -means with k equal to the true number of clusters, compute the NMI with the ground truth and the F-ratio test.
4. Apply the `sklearn` implementation of DBSCAN and MeanShift to the dataset. Discuss the differences among the obtained results.
5. Apply DP or other density based techniques and k -means to the dataset `spirals.txt` in the `Datasets` folder. The ground truth is available in `gt_spirals.txt`

Note

Hints for point 1. As a rule of thumb, ϵ should be chosen in such a way that, on average, between 1% and 2% of the data points are included in the neighbourhood. Use the exponential kernel.